5.4 Controlling network traffic to and from your virtual machine

You want traffic to enter or leave your EC2 instance only if it has to do so. With a firewall, you control ingoing (also called *inbound* or *ingress*) and outgoing (also called *outbound* or *egress*) traffic. If you run a web server, the only ports you need to open to the outside world are ports 80 for HTTP traffic and 443 for HTTPS traffic. All other ports should be closed down. You should only open ports that must be accessible, just as you grant only the permissions you need with IAM. If you are using a firewall that allows only legitimate traffic, you close a lot of possible security holes. You can also prevent yourself from human failure—for example, you prevent accidentally sending email to customers from a test system by not opening outgoing SMTP connections for test systems.

Before network traffic enters or leaves your EC2 instance, it goes through a firewall provided by AWS. The firewalls inspects the network traffic and uses rules to decide whether the traffic is allowed or denied.

IP vs. IP address

The abbreviation IP is used for Internet Protocol, whereas an IP address describes a specific address like 84.186.116.47.

Figure 5.8 shows how an SSH request from a source IP address 10.0.0.10 is inspected by the firewall and received by the destination